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Aufgabe 1

a)

Near Plane Projection

First we project the a point in eye space (x_e, y_e, z_e) onto the near plane (x_p, y_p, z_p)

$$x_p = \frac{n \cdot x_e}{-z_e}$$
$$y_p = \frac{n \cdot y_e}{-z_e}$$

Mirroring

As we divide in the step from clip coordinates to NDC by w_{clip} , we want to set this component to $-z_e$. Therefore the last row is constructed as follow

$$P = \begin{pmatrix} \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

Interpolation

Next we interpolate x from $[left, right]$ to $[-1, 1]$:

$$x_n = \frac{1 - (-1)}{right - left} \cdot x_p + \beta \quad \text{subst. } (x_n, x_p) \rightarrow (1, r)$$

$$1 = \frac{2 \cdot right}{right - left} + \beta$$

$$\beta = 1 - \frac{2 \cdot right}{right - left} = \frac{right - left}{right - left} - \frac{2 \cdot right}{right - left} = -\frac{right + left}{right - left}$$

$$\Rightarrow x_n = \frac{2 \cdot x_p}{right - left} - \frac{right + left}{right - left}$$

We follow the same procedure for y with $[bottom, top]$ to $[-1, 1]$

$$y_n = \frac{1 - (-1)}{top - bottom} \cdot y_p + \beta \Rightarrow y_n = \frac{2y_p}{top - bottom} - \frac{top + bottom}{top - bottom}$$

Substitute

We substitute x_e and y_e back into the equations

$$\begin{aligned} \Rightarrow x_n &= \frac{2 \cdot \frac{n \cdot x_e}{-z_e}}{right - left} - \frac{right + left}{right - left} = \frac{2n \cdot x_e}{(right - left) \cdot -z_e} - \frac{right + left}{right - left} \\ &\Rightarrow \left(\frac{2 \cdot near}{right - left} \cdot x_e + \frac{right + left}{right - left} \cdot z_e \right) / -z_e \end{aligned}$$

The last division is necessary to get the NDC, his step is performed by OpenGL.

$$\Rightarrow y_n = \frac{2 \cdot \frac{near \cdot y_e}{-z_e}}{top - bottom} - \frac{top + bottom}{top - bottom} = \left(\frac{2 \cdot near}{top - bottom} \cdot y_e + \frac{top + bottom}{top - bottom} \cdot z_e \right) /$$

We now have the following matrix:

$$P = \begin{pmatrix} \frac{2 \cdot near}{right - left} & 0 & \frac{right + left}{right - left} & 0 \\ 0 & \frac{2 \cdot near}{top - bottom} & \frac{top + bottom}{top - bottom} & 0 \\ \cdot & \cdot & \cdot & \cdot \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

Z-Projection

We want calculate the z-value based on the following formula:

$$z_n = \frac{A \cdot z_e + B}{-z_e}$$

To solve this, we will use the (z_e, z_n) Relation: $(-near, -1)$, $(-far, 1)$

$$\begin{cases} -A \cdot near + B = -near & (1) \\ -A \cdot far + B = far & (2) \end{cases}$$

$$(1') B = A \cdot near - near$$

substitute into (2)

$$-A \cdot far + A \cdot near - near = far \Rightarrow A = -\frac{far + near}{far - near}$$

Put A into (1) to find B

$$\frac{far + near}{far - near} \cdot near + B = -near \Rightarrow B = -\left(1 + \frac{far + near}{far - near}\right)near$$

$$\Rightarrow B = -\frac{2far \cdot near}{far - near}$$

We now have the complete matrix:

$$P = \begin{pmatrix} \frac{2 \cdot near}{right - left} & 0 & \frac{right + left}{right - left} & 0 \\ 0 & \frac{2 \cdot near}{top - bottom} & \frac{top + bottom}{top - bottom} & 0 \\ 0 & 0 & -\frac{far + near}{far - near} & -\frac{2far \cdot near}{far - near} \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

b)

To compute the inverse matrix we can follow a similar approach of **a)** and reverse each of these transformations. This results in the following matrix:

$$P = \begin{pmatrix} \frac{right - left}{2 \cdot near} & 0 & 0 & \frac{right + left}{2 \cdot near} \\ 0 & \frac{top - bottom}{2 \cdot near} & 0 & \frac{top + bottom}{2 \cdot near} \\ 0 & 0 & 0 & -1 \\ 0 & 0 & \frac{far - near}{2far \cdot near} & \frac{far + near}{2far \cdot near} \end{pmatrix}$$

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